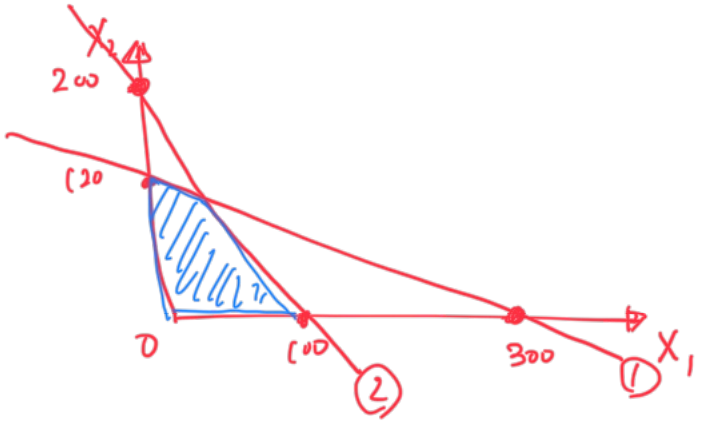
1. Consider the following production planning problem.

Maximize $Z=7X_1+5X_2$ (Profit, Unit: 10,000 KRW)
subject to
① $2X_1 + 5X_2 \leq 600$ (Labor, Unit: Hour)
② $4X_1 + 2X_2 \leq 400$ (Facility, Unit: Hour)
③④ $X_1, X_2 \geq 0$.

D.V 의사결정변수 X_1, X_2

(X_1: The production quantity of Product 1, X_2: The production quantity of Product 2)

(1-1) Graph the constraints and shade the feasible region.



$2X_1 + 5X_2 \leq 600$
↳ $2X_1 + 5X_2 = 600$
↳ 2개의 점

(1) $X_2 = 0$, $X_1 = ?$
$2X_1 + 5 \times 0 = 600$
↳ $2X_1 = 600$
↳ $X_1 = 300$ (X_1, X_2) $(300, 0)$

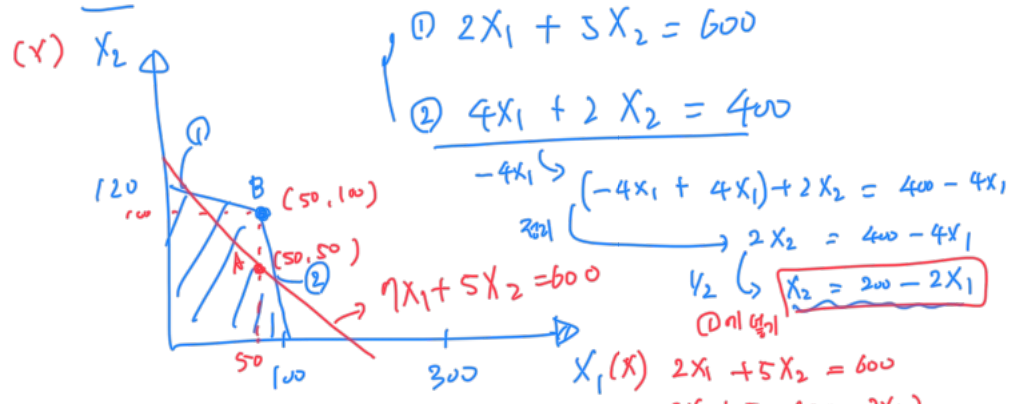
(2) $X_1 = 0$, $X_2 = ?$
$2 \times 0 + 5 \times X_2 = 600$
$5X_2 = 600$, $X_2 = 120$
$(0, 120)$

$4X_1 + 2X_2 \leq 400$
↳ $4X_1 + 2X_2 = 400$

(1) $X_2 = 0$, $X_1 = ?$
$4X_1 + 2 \times 0 = 4X_1 + 0 = 400$, $X_1 = 100$ $(100, 0)$

(2) $X_1 = 0$, $X_2 = ?$
$4 \times 0 + 2X_2 = 2X_2 = 400$, $X_2 = 200$ $(0, 200)$

(1-2) A = (50, 50) is a feasible solution that is not optimal. Explain, using the iso-profit line, how to find a feasible solution that is more profitable than A.



① $2X_1 + 5X_2 = 600$

② $4X_1 + 2X_2 = 400$

$-4X_1$ ↳ $(-4X_1 + 4X_1) + 2X_2 = 400 - 4X_1$

정리 ↳ $2X_2 = 400 - 4X_1$

$\frac{1}{2}$ ↳ $X_2 = 200 - 2X_1$

① 에 대입

$2X_1 + 5X_2 = 600$

$2X_1 + 5 \times (200 - 2X_1)$

$= 2X_1 + 5 \times 200 - 5 \times 2X_1$

$= 1000 - 8X_1 = 600$

↳ $1000 - 600 = 8X_1$

↳ $400 = 8X_1$

↳ $\frac{400}{8} = 50 = X_1$

$X_2 = 200 - 2X_1 = 200 - 2 \times 50 = 100$

$\therefore B(50, 100)$

o.f Maximize $Z = 7X_1 + 5X_2$

(1) $Z = 7 \times 50 + 50 \times 50 = 350 + 250 = 600$

(A) 제시된 o.f = $7X_1 + 5X_2 = 600$

$y = aX + b$ $X_2 = aX_1 + b$

$7X_1 + 5X_2 = 600$

$-7X_1$ ↳ $5X_2 = 600 - 7X_1$

$\frac{1}{5}$ ↳ $X_2 = \frac{600}{5} - \frac{7}{5}X_1$

↳ $2X_1 + 5X_2 = 600$

↳ $5X_2 = 600 - 2X_1$

↳ $X_2 = \frac{600}{5} - \frac{2}{5}X_1$

① $2X_1 + 5X_2 = 600 \rightarrow X_2 = \frac{600}{5} - \frac{2}{5}X_1$

② $4X_1 + 2X_2 = 400 \rightarrow X_2 = 200 - 2X_1$

$2X_2 = 400 - 4X_1$

↳ $X_2 = 200 - 2X_1$

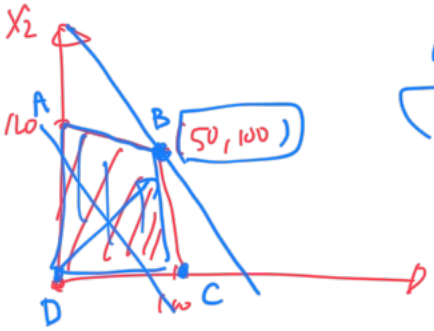
$X_2 = \frac{600}{5} - \frac{2}{5}X_1$

$X_2 = 200 - 2X_1$

$X_2 = \frac{600}{5} - \frac{7}{5}X_1$

$-2 < -\frac{7}{5} < -\frac{2}{5}$

(1-3) Find the optimal solution of this problem.



A (0, 120)

B (50, 100)

C (100, 0)

D (0, 0)

$Z = 7x_1 + 5x_2$

$Z = 7 \times 0 + 5 \times 120 = 600$

$Z = 7 \times 50 + 5 \times 100 = 850$

$C = 7 \times 100 + 5 \times 0 = 700$

$D = 7 \times 0 + 5 \times 0 = 0$

(1-4) Introduce a slack variable for each constraint and calculate their values at the optimal solution found in (1-3).

Capacity

(1) $2X_1 + 5X_2 = 600$

Amount used

$S_1 = 600 - (2X_1 + 5X_2)$

$= 600 - (2 \times 50 + 5 \times 100)$

$= 600 - (100 + 500)$

$= 0$

(2) $4X_1 + 2X_2 = 400$

$S_2 = 400 - (4X_1 + 2X_2)$

$= 400 - (4 \times 50 + 2 \times 100)$

$= 400 - (200 + 200)$

$= 0$

Ch $S_1 = 0$, $S_2 = 0$

(1-5) At the optimal solution, classify each constraint as binding or non-binding, and explain its managerial implication.

$S_1 = 0 \Rightarrow$ both binding.

$S_2 = 0 \Rightarrow$

① $2X_1 + 5X_2 \leq 600$ (Labor)

② $4X_1 + 2X_2 \leq 400$ (facility)

Labor 600 넘나 크게 and/or Facility 400 넘나 크게

증가시킬 수 있다.

2. Solve each of the following linear optimization models graphically. For each model, find the optimal solution, calculate the slack variables, identify the binding and non-binding constraints, and explain their managerial implications.

(2-1) Maximize $X_1 + X_2$

subject to

$2X_1 + X_2 \leq 10$

$X_1 + 3X_2 \leq 15$

$X_1 \geq 0, X_2 \geq 0$

(2-2) Maximize $X_1 + 4X_2$

subject to

$$X_1 + 2X_2 \leq 8$$
$$3X_1 + 2X_2 \leq 12$$
$$X_1 \geq 0, \quad X_2 \geq 0$$